

Thrombotic Microangiopathies

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INTRODUCTION

Thrombotic microangiopathies (TMAs) are the consequence of severe endothelial injury with pathological features representing the tissue response to injury.¹ It can manifest in a diverse range of conditions and presentations, but acute kidney injury (AKI) is a common prominent feature because of the propensity of the glomerular circulation to endothelial damage and occlusion.

CLASSIFICATION

Over the past 20 years the classification of TMAs has evolved to reflect our increasing understanding of the pathophysiology of disease. Historically, classifications reflected the predominant phenotype: hemolytic uremic syndrome (HUS) for renal dominant disease and thrombotic thrombocytopenic purpura (TTP) for predominant neurological involvement. More recently, severe ADAMTS13 deficiency defined TTP; the

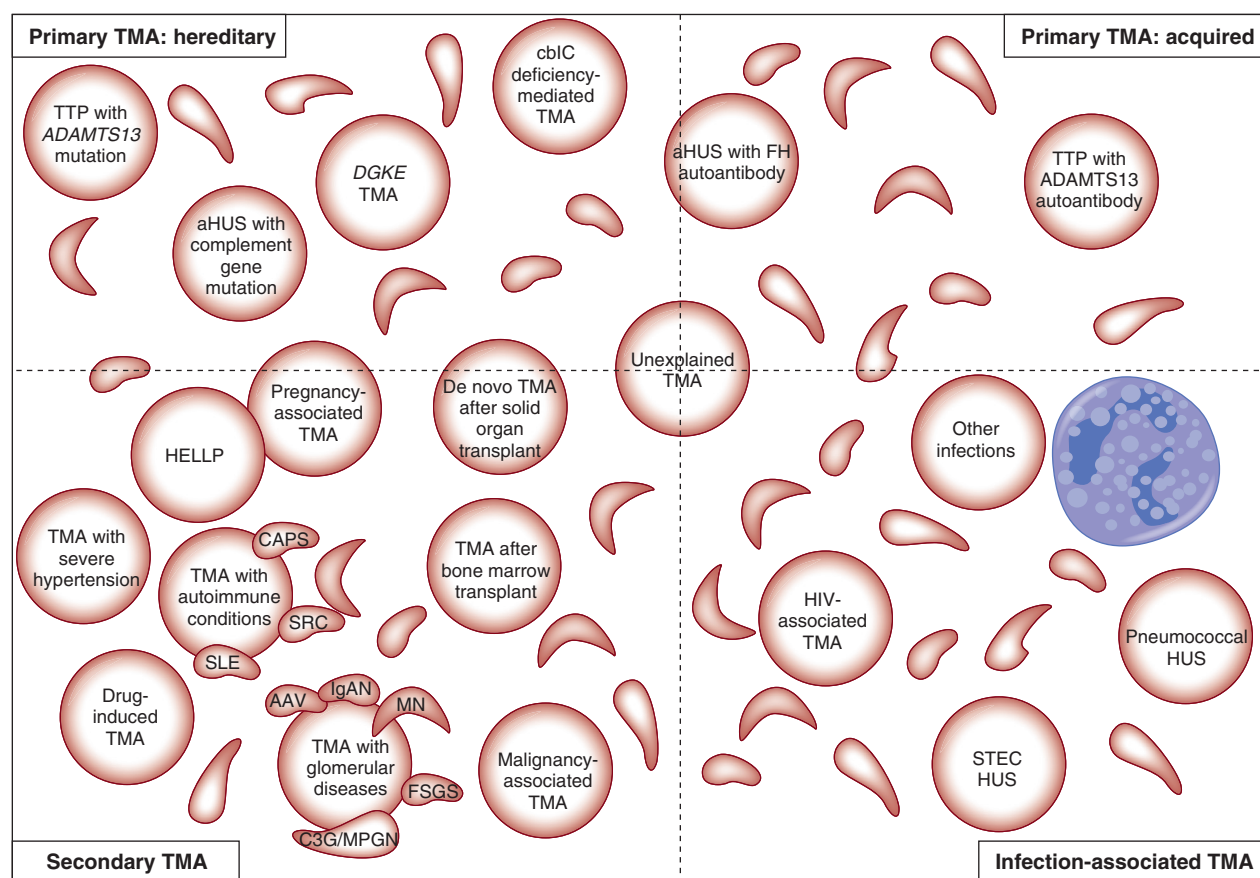


Fig. 34.1 Classification of thrombotic microangiopathies (TMAs). TMAs are classified as primary inherited TMAs, primary acquired TMAs, infection-associated TMAs, and secondary TMAs. Primary inherited TMAs are predisposed to by mutations in *ADAMTS13*, *MMACHC*, or genes encoding complement proteins. Primary acquired TMAs are caused by autoantibodies to *ADAMTS13* or complement *FH*. Infection-associated TMAs may have distinct mechanisms (e.g., Shiga toxin-producing *Escherichia coli* and pneumococcal hemolytic uremic syndrome). In other infections the processes are not defined and in some cases the infection may trigger manifestation of a primary TMA. Secondary TMAs occur in a spectrum of conditions, and in many cases the pathogenic mechanisms are multifactorial or unknown. In some secondary cases (e.g., pregnancy-associated TMA or de novo TMA after transplantation), a significant proportion of individuals will have a genetic predisposition to a primary TMA. AAV, ANCA (antineutrophil cytoplasmic antibody)-associated vasculitis; *ADAMTS13*, a disintegrin and metalloproteinase with a thrombospondin type 1 motif, member 13; *aHUS*, atypical hemolytic uremic syndrome; *C3G*, C3 glomerulopathy; *CAPS*, catastrophic antiphospholipid syndrome; *cbIC*, cobalamin C type; *DGKE*, gene encoding diacylglycerol kinase ϵ ; *FH*, factor H; *FSGS*, focal segmental glomerulosclerosis; *HELLP*, syndrome of hemolysis, elevated liver enzymes, and low platelets; *HIV*, human immunodeficiency virus; *IgAN*, IgA nephropathy; *MN*, membranous nephropathy; *MPGN*, membranoproliferative glomerulonephritis; *SLE*, systemic lupus erythematosus; *SRC*, scleroderma renal crisis; *TMA*, thrombotic microangiopathy; *TTP*, thrombotic thrombocytopenic purpura. (Modified from the National Renal Complement Therapeutics Centre 2017/2018 annual report. Available from: <http://www.atypicalhus.co.uk/>. Last accessed February 4, 2019.)

presence of Shiga toxin-producing bacteria categorized Shiga toxin-associated HUS (STEC-HUS) and broadly all other causes of TMA could be classified as atypical HUS (aHUS). Current classifications designate primary inherited TMAs (e.g., complement mutations, *ADAMTS13* mutations), primary acquired TMAs (factor H autoantibodies, *ADAMTS13* autoantibodies), infection-associated TMAs, and secondary TMAs (Fig. 34.1).¹ It is, however, necessary to remember that patients with an underlying complement genetic mutation often require a secondary trigger for aHUS to manifest.

CLINICAL AND LABORATORY FEATURES

The clinical features of TMA reflect hemolysis and ischemic organ dysfunction and the underlying cause. Specific features of different etiologies are covered in individual sections.

Thrombocytopenia and microangiopathic hemolytic anemia (MAHA) are defining features of TMAs. Aggregation and consumption account for the low platelet count, and the MAHA is thought to be caused by mechanical fragmentation in areas of disturbed flow in small blood vessels due to partial occlusion by thrombus (Fig. 34.2A). An elevated lactate dehydrogenase is the most sensitive marker of hemolysis and is due to cell lysis and tissue ischemia. Haptoglobin levels are low in TMAs due to the binding of free hemoglobin and the uptake of haptoglobin-hemoglobin complexes in the reticuloendothelial system. A direct antiglobulin test (DAT/direct Coombs) is negative with the exception of pneumococcal HUS. AKI is a common manifestation of TMAs due to ischemia in the kidney, although rarely a severe feature of TTP. Once routine hematological and biochemical diagnostics have confirmed a TMA, investigations are aimed at determining the underlying etiology and excluding other differential

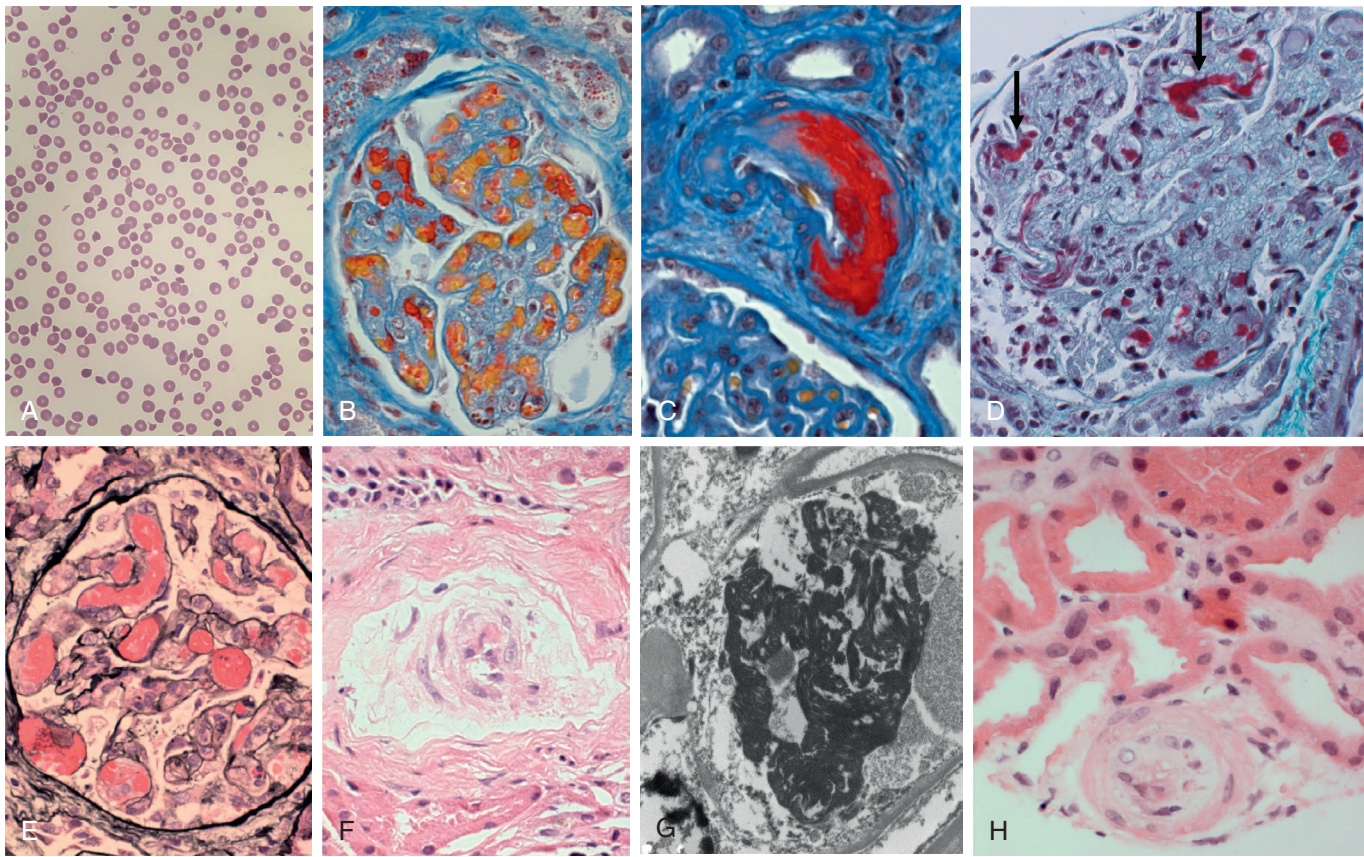


Fig. 34.2 The pathological features of thrombotic microangiopathies. (A) Blood film of a patient with thrombotic microangiopathy demonstrating the presence of schistocytes. (B) Fibrin thrombi (red) and erythrocytes (yellow) in glomerular capillary lumina (*Martius Scarlet Blue*). (C) Fibrinoid necrosis of arterial wall (*Martius Scarlet Blue*). (D) Thrombi (arrows) in capillary loops (*Masson trichrome*). (E) Glomerular paralysis with capillary loops containing abundant erythrocytes (silver). (F) Myxoid intimal thickening of small artery (*hematoxylin and eosin*). (G) Fibrin tactoids (black) in glomerular capillary (electron micrograph). (H) Muroid thickening and obliteration of the lumen of a small artery (*hematoxylin and eosin*).

diagnoses—the most urgent test being an ADAMTS13 assay (Fig. 34.3).¹

THROMBOTIC MICROANGIOPATHY PATHOLOGY

In acute HUS, glomerular capillary wall thickening is seen as a result of endothelial cell swelling and accumulation of flocculent material between the endothelial cell and the basement membrane. Fibrin and platelet thrombi may be seen, resulting in occlusion of the glomerular capillaries and glomerular paralysis. It has been noted that overt fibrin platelet thrombosis may be absent from renal biopsies of TMA, which has recently led to a suggested reclassification to microangiopathy with or without thrombosis.² Mesangiolysis occurs early in the disease process and subsequently is replaced by sclerotic changes. Early arterial changes are variable, ranging from only mild endothelial swelling to fibrinoid necrosis with occlusive thrombus formation. Erythrocyte fragments may be seen in the vessel wall. Subsequently, there is muroid intimal hyperplasia with narrowing of the vessel lumen (Fig. 34.2).

Deposition of fibrin or fibrinogen in the glomeruli and in the mesangium, as well as within the vessel walls, is seen

on immunofluorescence. Complement and immunoglobulin deposits along the capillary loops of glomeruli may be seen.

Evidence of TMA has also been reported in a number of glomerular diseases and autoimmune diseases; however, in published clinicopathologic studies only a small proportion of individuals with pathological evidence of TMA in these contexts had concurrent clinical and laboratory evidence. It is not possible based on current knowledge to establish TMA etiology from the histopathological morphology, though this may evolve with further research.

TTP is characterized by unusually large multimers of von Willebrand factor (vWF) and platelet-rich thrombi in capillaries and arterioles, although with current practice pathological specimens are rarely available.

PRIMARY THROMBOTIC MICROANGIOPATHIES

COMPLEMENT-MEDIATED ATYPICAL HEMOLYTIC UREMIC SYNDROME

A series of ground-breaking studies in the late 1990s established the role of complement overactivation in the pathogenesis of aHUS.³

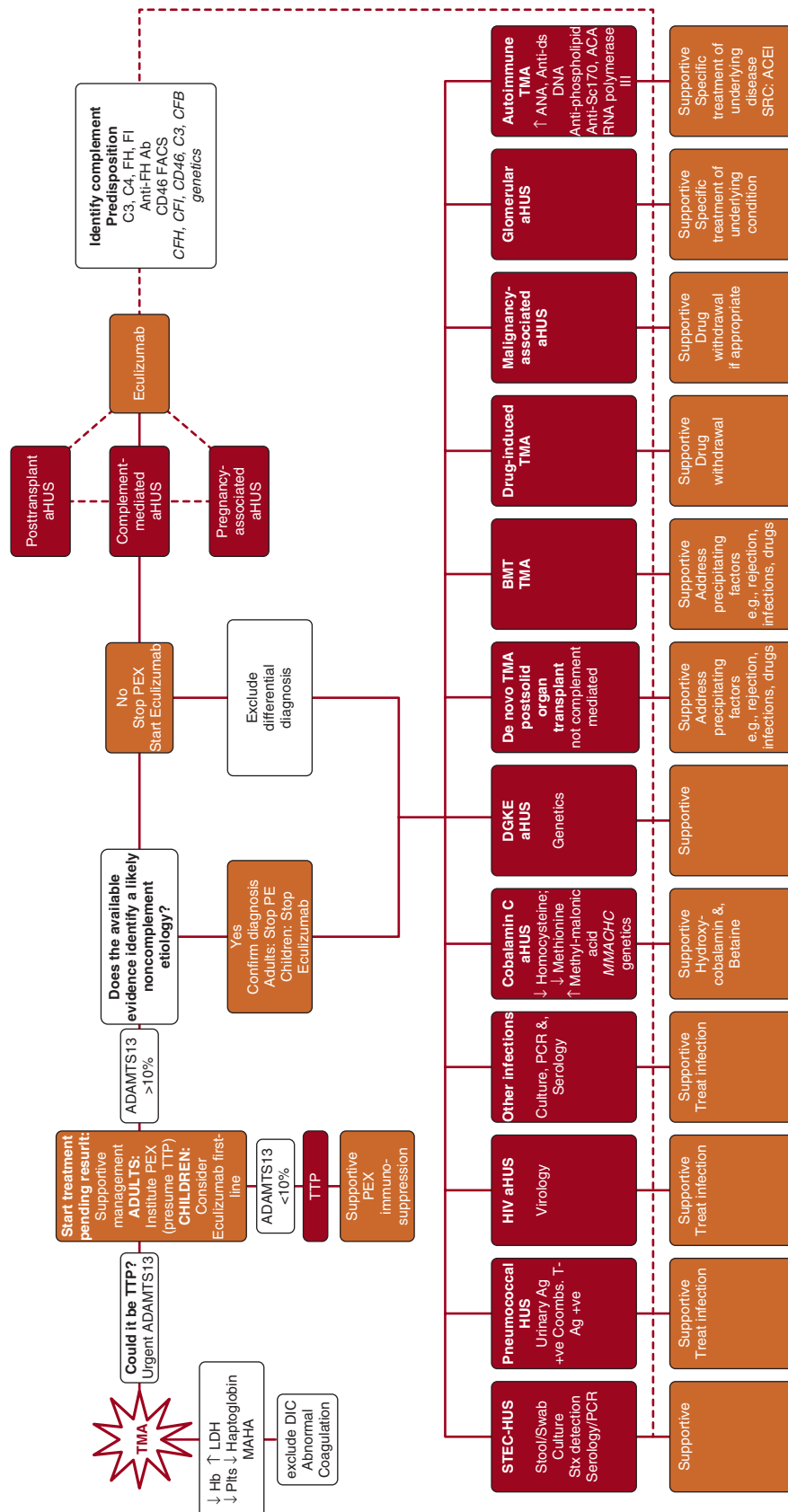


Fig. 34.3 Algorithm for the investigation and management of a patient presenting with thrombotic microangiopathy. Urgent treatment is critical given the high mortality if untreated. In adults, plasma exchange (PEX) should be instituted on the presumption that it is thrombotic microangiopathy (TMA) unless strong evidence suggests otherwise. In children, TTP is rarer and initial treatment with eculizumab should be considered if complement-mediated atypical hemolytic uremic syndrome (aHUS) is suspected and should not be delayed while ADAMTS13 activity is determined. There is no rapid diagnostic test which identifies complement-mediated aHUS in the acute phase and in those with ADAMTS13 activity greater than 10% without an obvious other etiology, eculizumab should be commenced. A diagnostic screen for noncomplement etiologies should then be recommended to determine optimal long-term treatment. Where eculizumab is not available, PEX should be continued until remission is established. Complement evaluation is recommended in all cases due to secondary causes of TMA unmasking latent complement defects. *BMT*, Bone marrow transplant; *DGKE*, diacylglycerol kinase ϵ ; *DIC*, disseminated intravascular coagulation; *FACS*, fluorescence-activated cell sorting; *FH*, factor H; *FI*, factor I; *Hb*, hemoglobin; *HIV*, human immunodeficiency virus; *LDH*, lactate dehydrogenase; *MAHA*, microangiopathic hemolytic anemia; *MMACHC*, methylmalonic aciduria and homocystinuria type C protein; *PCR*, polymerase chain reaction; *Pits*, platelets; *STEC-HUS*, Shiga-toxin-producing *Escherichia coli*-HUS; *TMA*, thrombotic microangiopathy.

The complement system comprises around 50 plasma and cell-surface proteins operating in a regulated network of signaling and amplification central to the normal physiological functioning of both innate and adaptive immune systems. Equal numbers of complement activators and inhibitors exist. In addition to opsonization and lysis of pathogens and modulation of the adaptive immune system, complement also facilitates the disposal of damaged host cells and potentially damaging immune complexes.

Complement activation can be initiated by three pathways: classical pathway (CP), lectin pathway (LP), and alternative pathway (AP; Fig. 34.4). The CP is triggered when C1q binds to one of a number of activators such as antigen–antibody complexes and other nonimmunoglobulin moieties. C1q activates the associated serine proteases, C1r and C1s, which cleave the plasma proteins C2 and C4 to form a C3 convertase complex (C4b2a) on the activating surface. The LP is activated when microbial carbohydrates are recognized by mannose-binding lectin (MBL), ficolins, or collectins and induces MBL-associated serine proteases to cleave C2/C4, once again forming C4b2a. The AP is constitutively active at a low level (tick over), but is primarily responsible for the rapid amplification of the CP and LP. The C3 convertases activate C3, resulting in the deposition

of C3b and the release of C3a. Through interplay with the soluble proteases factor B and D (FB and FD, respectively), surface-bound C3b forms additional C3 convertases. Thus the alternate pathway is responsible for approximately 80% of complement activity even when initially triggered through the CP.

The terminal pathway of complement begins when continuous deposition of C3b and stabilization of convertases by properdin lead to the formation of C5 convertases, which cleave C5 to the anaphylatoxins C5a and C5b that after a conformational change bind C6, C7, C8, and C9 to form the membrane attack complexes that induce lysis or damage of targeted cells (Fig. 34.4).

COMPLEMENT REGULATION

The amplification loop of the complement system provides a rapid response to pathogens but leaves host cells susceptible to collateral damage if the complement system is unchecked. The system is, therefore, tightly regulated by plasma proteins, including factor H and factor I and cell surface proteins such as CD46. Impaired regulation can lead to pathology and complement-mediated aHUS is the archetypal disease of excessive complement activation (Fig. 34.4).

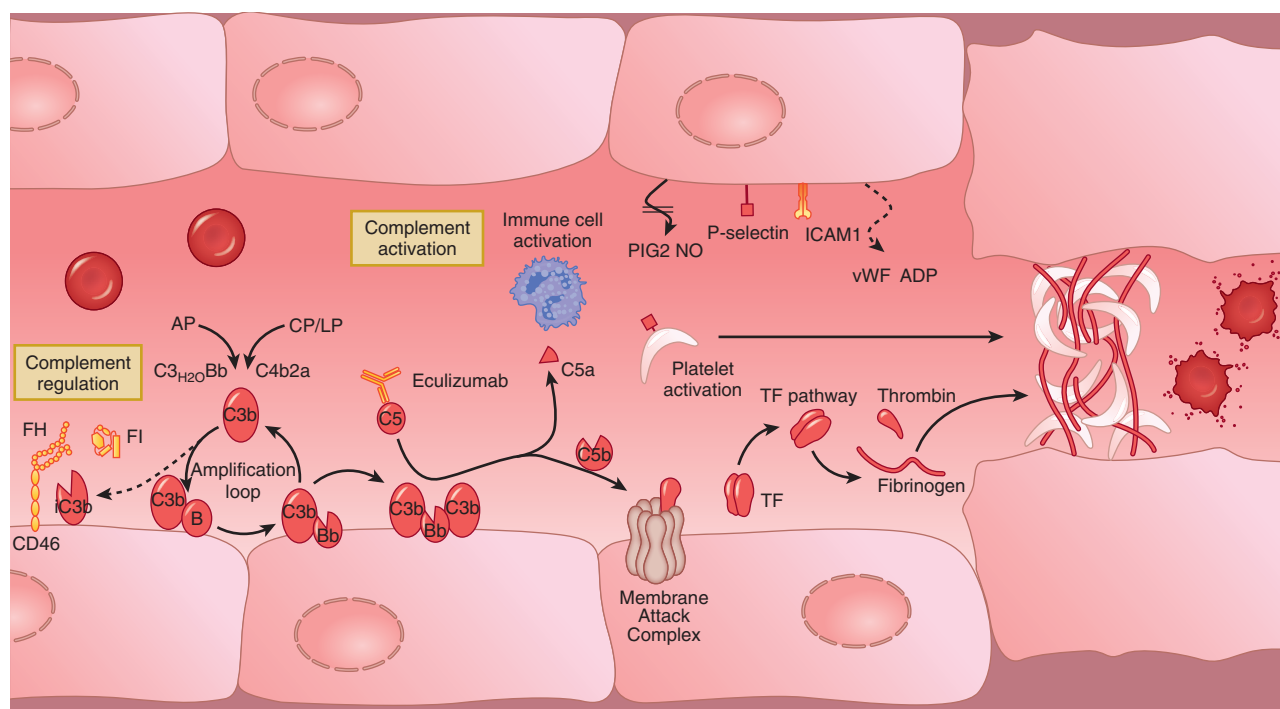


Fig. 34.4 Complement-mediated atypical hemolytic uremic syndrome (aHUS). Complement is activated by the alternative (AP), classical (CP), and lectin (LP) pathways. The AP is a positive amplification loop. C3b interacts with factor B, which is then cleaved by factor D to form the C3 convertase C3bBb. Unfettered, this leads to activation of the terminal complement pathway with generation of the anaphylatoxin C5a and the membrane attack complex (MAC, C5b-9). The glomerular endothelium is protected from collateral damage from the AP by complement regulators including factor H (FH), factor I (FI), and CD46. In complement-mediated aHUS, loss-of-function mutations in *CFH*, *CFI*, and *CD46*, activating mutations in *C3* and *CFB*, and autoantibodies to FH result in overactivation of the AP. This leads to immune cell and platelet activation and endothelial cell damage and swelling, with consequent thrombus formation, platelet consumption, vascular occlusion, and mechanical hemolysis. Eculizumab binds to C5 to prevent activation of the terminal pathway inhibiting the generation of the effector molecules that cause thrombotic microangiopathy (TMA) in individuals in whom a primary defect in complement underlies the TMA pathogenesis. (Modified from the National Renal Complement Therapeutics Centre 2017/2018 annual report. Available from: <http://www.atypicalhus.co.uk/>. Last accessed February 4, 2019.)

INHERITED COMPLEMENT-MEDIATED ATYPICAL HEMOLYTIC UREMIC SYNDROME

Complement-mediated aHUS is a rare disease with an incidence of about 0.4 per million population per year.

COMPLEMENT FACTOR H

Genetic studies in 1998 first established that loss-of-function mutations in factor H (*CFH*) caused complement-mediated aHUS.⁴ Since then, multiple studies have confirmed that mutations in this gene are the most common cause of disease, accounting for around 25% of cases.

The factor H protein (FH) is the most important fluid-phase regulator of the AP of complement. It consists of 20 modules called complement control proteins modules (CCPs). The four N-terminal CCPs (CCPs 1–4) mediate the complement regulatory functions of FH by (1) competing with FB for C3b binding, (2) accelerating the decay of the C3 convertase into its components, and (3) acting as a cofactor for factor I-mediated proteolytic inactivation of C3b. The C-terminal domains of FH (CCP19–20) mediate host surface protection by binding polyanions such as glycosaminoglycans. It is this C-terminal region of FH that is most commonly mutated in aHUS, resulting in impaired cell-surface complement regulation.⁵

It is the genetic architecture of the regulators of complement activation (RCA) gene cluster on chromosome 1, in which *CFH* resides, which is responsible for this mutational hotspot. The RCA cluster is thought to have arisen from several large genomic duplications and *CFH* and five factor H-related proteins thus have a very high degree of sequence identity. This homology predisposes to gene conversions and genomic rearrangements (*CFH/CFHR1* and *CFH/CFHR3* hybrid genes) through nonallelic homologous recombination and microhomology-mediated end joining.⁶ These hybrid genes, as with C-terminal point mutations in *CFH*, result in loss-of-cell-surface complement regulation. From a practical perspective, this complex genomic locus is very hard to sequence when compared with other genes mutated in human disease, in part, because of large regions of repetitive genomic DNA sequences.

More recently a reverse *CFHR1/CFH* hybrid gene arising through nonallelic homologous recombination in which the C-terminal CCPs of FHR1 are replaced by the C-terminal CCPs of FH has been described. In this setting this FHR1/FH hybrid protein does not impair FH cell surface binding but instead, acts as a competitive inhibitor of FH.⁷

CD46 MEMBRANE COFACTOR PROTEIN

Mutations in *CD46* (membrane cofactor protein) are found in approximately 10% of patients with aHUS. CD46 is a membrane inhibitor of complement activation expressed on most cell types with the exception of erythrocytes. It consists of four CCP modules containing the sites for complement interaction followed by an O-glycosylation region, enriched with serine, threonine, and proline residues (STP domain); a transmembrane anchor; and a cytoplasmic tail. Most mutations described in aHUS are found in the

extracellular four CCP domains that are responsible for C3b and C4b binding. The majority of *CD46* mutations resulted in reduced cell surface expression of CD46 (75%) with the remaining producing a surface expressed but nonfunctional protein.⁸

COMPLEMENT FACTOR I

Mutations in complement factor I (*CFI*) account for between 5% and 10% of aHUS.⁹ FI is a disulfide-linked heterodimer serine protease that cleaves C3b and C4b in conjunction with a cofactor protein (FH for C3b; C4b binding protein for C4b; CD46 and complement receptor 1 for both). The *CFI* mutations described in aHUS are all heterozygous.³

COMPLEMENT C3

C3 is the nexus of the complement cascade. It is cleaved to form the anaphylatoxin C3a and the highly reactive C3b which can then bind to cell surfaces via its reactive thioester. Thioesters are compounds with the functional group R–S–CO–R'. C3b can then interact with FB in the presence of FD to form the AP convertase. Mutations in C3 account for around 2% to 8% of complement-mediated aHUS.¹⁰ They represent gain-of-function mutations. The mutations in C3 linked to complement-mediated aHUS result in complement overactivation by either binding to FB with greater affinity or preventing complement regulators binding to C3 and inactivating it. Most C3 mutations are associated with low serum C3 levels (70%–83%).

COMPLEMENT FACTOR B

Gain-of-function mutations also have been reported in *CFB*, although these appear to be rare.¹¹ FB carries the catalytic site of the complement AP convertase (C3bBb). Complement overactivation occurs by impaired complement regulation or increased convertase formation.

THROMBOMODULIN

Genetic variants in thrombomodulin (*THBD*) have been reported in some aHUS cohorts, however their causality remains to be established.¹² THBD regulates clot formation by the activation of protein C by thrombin and enhancing thrombin-mediated activation of plasma procarboxypeptidase B (CPB2), an inhibitor of fibrinolysis. THBD, however, has been suggested to have a role in the regulation of the AP by accelerating FI-mediated inactivation of C3b.¹²

DISEASE PENETRANCE

In complement-mediated aHUS, genetic mutations are not causative per se but are instead predisposing. For a disease to manifest, additional genetic and environmental modifiers are required (Fig. 34.5). The disease penetrance is age related and has been reported to be as high as 64% by the age of 70 years in individuals with a single rare genetic variant, although this is likely to represent the upper range. Rarely aHUS patients (about 3%) will have more than one mutation with increased penetrance per additional mutation.

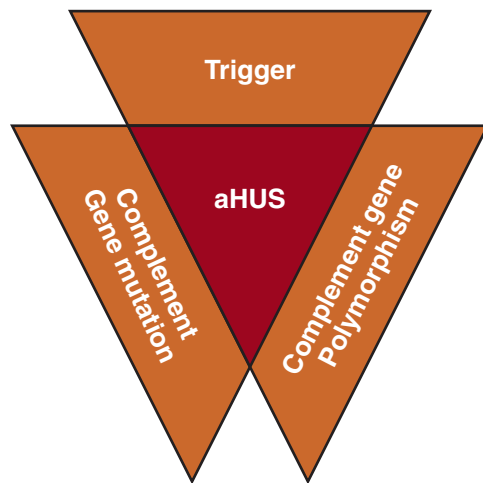


Fig. 34.5 Multihit model of complement-mediated atypical hemolytic uremic syndrome (aHUS). For complement-mediated aHUS to manifest, in addition to a rare genetic mutation in a complement gene, an individual may require common genetic variants such as an at-risk haplotype in a complement gene and the presence of a trigger (e.g., pregnancy, infection). (Modified from the National Renal Complement Therapeutics Centre 2017/2018 annual report. Available from: <http://www.atypicalhus.co.uk/>. Last accessed February 4, 2019.)

In addition to rare genetic variants, common genetic single-nucleotide polymorphisms (SNPs) have been associated with complement-mediated aHUS. A haplotype block in encompassing the *CFH* gene (*CFH*-H3; tgtgt) composed of these SNPs increases this risk of aHUS twofold to fourfold. Functional analysis has shown that the risk variant, *CFH*Val62, has a subtle decrease in cofactor activity compared with the protective variant. Another haplotype block in the *RCA* cluster, *CD46*^{ggaac}, has been associated with two- to threefold increased risk of aHUS, although the functional significance has yet to be established. Even when there are multiple rare and common genetic risk factors for complement-mediated aHUS, a trigger is frequently required (e.g., pregnancy, infection) to unmask the latent complement defects and cause aHUS. These triggers usually activate complement. For instance, complement activation is the normal physiological response to infection.³ In some cases of secondary TMA a high proportion of individuals carry a mutation (e.g., pregnancy-associated aHUS, approximately 70%, and de novo posttransplant TMA, approximately 30%) but in others the incidence of mutations is unknown or low (e.g., STEC-HUS). In other TMAs, complement activation may be seen in vivo but it is yet to be determined if in these settings complement is causative, a modifier, or simply a bystander (Fig. 34.6).

ACQUIRED PRIMARY COMPLEMENT-MEDIATED ATYPICAL HEMOLYTIC UREMIC SYNDROME

In addition to inherited defects in the complement system, acquired defects in the form of autoantibodies to FH¹³ and FI¹⁴ have been reported in complement-mediated aHUS. FH autoantibody aHUS predominantly presents in childhood and in European cohorts accounts for 5% to

25% of cases, although in one Indian cohort this was as high as 56%.

There is a genetic predisposition to FH autoantibodies in aHUS with FHR1 deficiency either by way of whole gene deletions of *CFHR3* and *CFHR1* or *CFHR1* and *CFHR4* or by point mutations in the *CFHR1* gene strongly linked. The role of FHR1 deficiency in the generation of FH autoantibodies is obscure and several patients have been reported with factor H autoantibodies in the presence of FHR1.¹⁵

The FH autoantibodies predominantly bind to C-terminal epitopes on FH and perturb FH-mediated cell-surface protection. Thus these acquired defects mirror the genetic defects seen in FH-associated aHUS.

Autoantibodies against FI have also been reported but are rare and their functional relevance remains to be established.

COMPLEMENT SCREENING IN ATYPICAL HEMOLYTIC UREMIC SYNDROME

Complement analysis in cases of aHUS should include serum levels of C3, C4, FH, FI, FB, FBb, and sC5b-9 and factor H autoantibodies prior to plasma exchange (PEX; Fig. 34.3).² Low C3 levels are commonly seen in patients with complement mutations and may point to a complement-mediated process; however, normal C3 levels do not exclude the presence of a mutation or autoantibodies. Flow cytometry of peripheral blood mononuclear cells to detect reduced CD46 is also advised. Mutation screening in aHUS is challenging, because most of the disease-associated mutations are individually rare. In the case of nonsense mutations, large gene rearrangements, and frame shift mutations, the functional consequences are clear. In complement-mediated aHUS, a significant proportion of variants consist of missense mutations of unknown significance which pose a challenge when reporting the genetic results. In addition to direct DNA sequencing, the high frequency of gene conversions and genomic rearrangements found in aHUS makes copy number analysis essential in aHUS screening. A polymorphism in the *C5* gene (p.R885H) has been demonstrated to prevent eculizumab binding to C5 and genetic screening should be undertaken to identify those that will not respond to eculizumab therapy, allowing early reinitiation of PEX.³

CLINICAL FEATURES OF COMPLEMENT-MEDIATED ATYPICAL HEMOLYTIC UREMIC SYNDROME

AKI is a predominant feature of complement-mediated aHUS.¹ Historically the prognosis for patients with aHUS was poor. Initial mortality was been reported to be higher in children (6.7% vs. 0.8% at 1 year), although adults progress to end-stage renal disease (ESRD) more frequently at initial presentation (46% vs. 16%). At 3 to 5 years after onset, 64% to 67% of adults and 36% to 48% of children had died or reached end-stage renal failure (ESRF). Prognosis varies according to genotype, with *CD46* mutations carrying the best prognosis. No patient with a *CD46* mutation from two large reported cohorts died at first episode and no children and only 25% of adults reached ESRF at first presentation. At 3

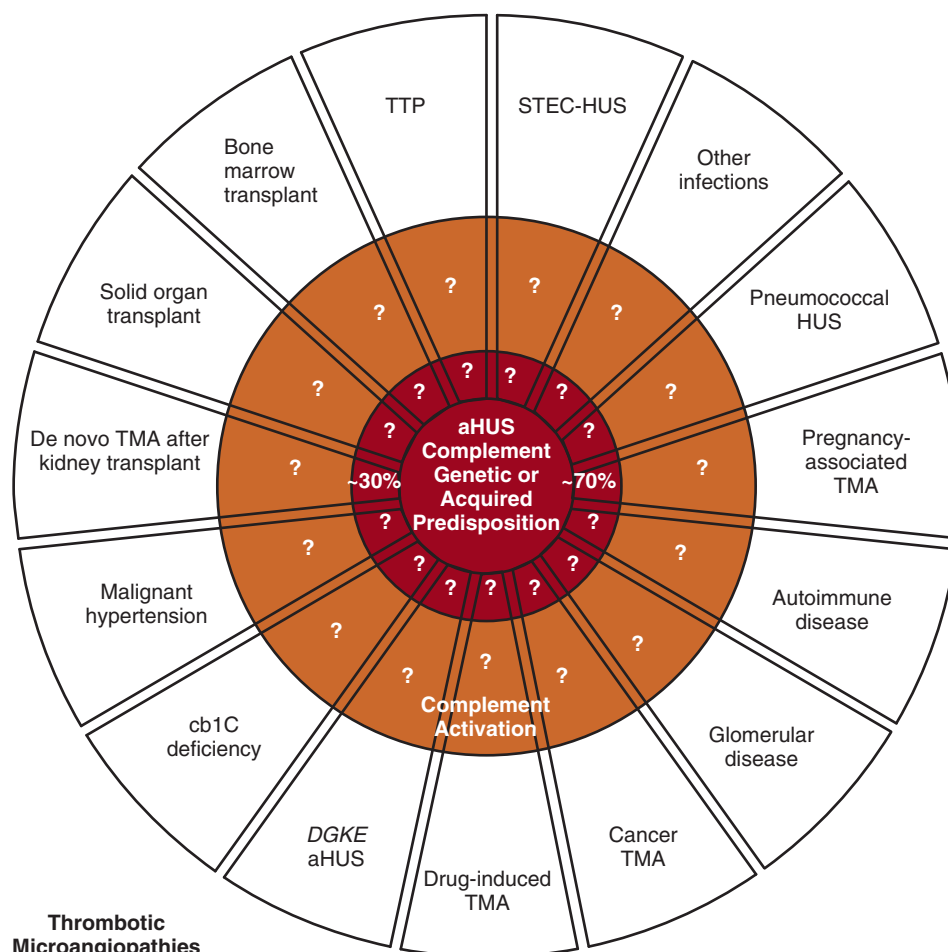


Fig. 34.6 The role of complement in thrombotic microangiopathies (TMAs). In primary inherited and acquired complement-mediated atypical hemolytic uremic syndrome (aHUS) a mutation or autoantibody resulting in complement dysregulation predisposes to disease. Complement-mediated aHUS frequently only manifests upon exposure to an environmental trigger, which can include other causes of TMA. In some cases of secondary TMA a high proportion of individuals carry a mutation (e.g., pregnancy-associated aHUS, about 70%, and de novo posttransplant TMA, about 30%), but in others the incidence of mutations is unknown or low (e.g., STEC-HUS). In other TMAs, complement activation may be seen in vivo but it is yet to be determined if in these settings complement is causative, a modifier, or simply a bystander. STEC-HUS, Shiga toxin-producing *Escherichia coli*-hemolytic uremic syndrome; TTP, thrombotic thrombocytopenic purpura. (Modified from the National Renal Complement Therapeutics Centre 2017/2018 annual report. Available from: <http://www.atypicalhus.co.uk/>. Last accessed February 4, 2019.)

years following initial presentation only 6% of both adults and children had developed ESRF. By contrast, individuals with mutations in *CFH*, *CFI*, or *C3* all had poor outcomes. In those with *CFH* mutations initial mortality was 30% in children and 4% in adults and evolution to ESRD at first episode in survivors was 19% to 33% in children and 48% in adults. At 5 years up to 77% of patients with *CFH* mutations had developed ESRD or had died. Only 30% to 40% of individuals with *CFI* and *C3* mutations will be alive with native kidney function at 3 to 5 years. A small proportion of patients will have multiple rare genetic variants. In those with CD46 mutations additional mutations worsened the prognosis. In those with FH autoantibody-mediated aHUS 36.5% to 63% died or reached ESRD by 5 years. Eculizumab is a recombinant humanized monoclonal antibody directed against the C5 complement protein. Eculizumab inhibits the cleavage of C5 by C5 convertase. The introduction of eculizumab has altered the natural history of disease and the outcome for those with complement abnormalities is markedly improved.

In complement-mediated aHUS extrarenal manifestations are reported in only 10% to 20% of patients with aHUS (Box 34.1). Neurological sequelae are the commonest reported symptom with symptoms ranging from irritability to coma. It is unclear whether these clinical events are a direct consequence of the TMA, a direct effect of complement activation, or complications of AKI, such as severe hypertension and uremia.³

TREATMENT OF COMPLEMENT-MEDIATED ATYPICAL HEMOLYTIC UREMIC SYNDROME

PLASMA EXCHANGE

PEX is currently recommended in TMA until an ADAMTS13 activity is available to exclude TTP. In addition to removing ADAMTS13 autoantibodies and replacing ADAMTS13, PEX will replace nonfunctioning complement proteins and remove

Box 34.1 Extrarenal Manifestation in Complement-Mediated Atypical Hemolytic Uremic Syndrome and Shiga Toxin–Producing *Escherichia Coli*-Hemolytic Uremic Syndrome

- Neurologic involvement, including seizures and altered consciousness
- Pancreatitis
- Diabetes mellitus
- Cardiac involvement/myocardial infarction
- Gastrointestinal involvement (including diarrhea, vomiting, abdominal pain)
- Ocular involvement
- Digital gangrene/skin
- Cerebral artery thrombosis/stenosis
- Extracerebral artery stenosis
- Pulmonary involvement
- Hepatitis
- Hemorrhagic colitis
- Bowel necrosis
- Bowel perforation
- Intussusception
- Colonic strictures

FH-autoantibodies and hyperfunctional complement components. Once TTP has been excluded eculizumab should be used, although its current high price means that PEX will remain the only option in many countries. Consensus guidelines recommend using 1–2 plasma volumes per session in adults and 50–100 mL/kg in children. PEX is initially performed daily with a reduced frequency when hemolysis is controlled. In those with complement defects plasma dependency was frequently seen and some patients required long-term plasma therapy (weekly/biweekly) to maintain remission.

ECULIZUMAB

The discovery of the role of complement in disease provided the rationale for the use of eculizumab.² Eculizumab is a recombinant humanized monoclonal antibody that binds C5 and blocks its subsequent cleavage into C5a and C5b. Eculizumab appears highly effective, with approximately 85% of patients becoming disease free in both plasma-resistant and plasma-dependent aHUS.¹⁶ No biomarker currently exists that will diagnose a primary complement-mediated aHUS in the acute setting, and the diagnosis is one of exclusion. As early initiation of eculizumab has been shown to lead to better outcomes, treatment is often commenced in patients with suspected primary complement-mediated aHUS, and discontinued if an alternative etiology is subsequently identified (Fig. 34.3).¹

The optimal treatment duration in complement-mediated aHUS is uncertain, although currently the license is for long-term treatment. Eculizumab has been withdrawn in a large series of patients with complement-mediated aHUS with relapse reported in about 33% of patients. All

individuals who relapsed had rare genetic variants in the complement system but importantly, rapid reintroduction of complement inhibition normalized the renal function. Prospective trials of disease-driven intermittent eculizumab regime are underway to define the optimal treatment duration.

ECULIZUMAB NONRESPONSIVE ATYPICAL HEMOLYTIC UREMIC SYNDROME

With increasing use in clinical practice it has become apparent that subgroups of patients with aHUS do not respond to eculizumab.¹⁷ In an industry study, all pediatric patients with a rare complement genetic variant or an autoantibody to FH had an improvement in estimated glomerular filtration rate, whereas 27% of individuals without an identified complement abnormality failed to show an improvement. It is not clear whether this represents late presentation of disease or true nonresponse. In addition, individuals with methylmalonic aciduria and homocystinuria type C protein (MMACHC)–,¹⁸ diacylglycerol kinase ϵ (DGKE)–,¹⁹ and inverted formin 2 (INF2)–mediated²⁰ aHUS have been shown to fail to respond to treatment. Individuals with the C5 polymorphism (p.R885H) will also fail to respond to treatment.²¹

ECULIZUMAB SIDE EFFECTS

Susceptibility to infection with encapsulated gram-negative organisms, particularly *Neisseria* infections, is the most serious side effect of eculizumab treatment. Because of this, meningococcal vaccination is mandatory although its effectiveness in the context of complement blockade is uncertain. Long-term antibiotic prophylaxis is thus recommended, although meningococcal infection can occur despite these precautions and patient awareness is crucial.² Hepatotoxicity and deposition of eculizumab in the glomerulus have been reported.

RENAL TRANSPLANTATION IN COMPLEMENT-MEDIATED ATYPICAL HEMOLYTIC UREMIC SYNDROME

Historically the outcome of renal transplantation in complement-mediated aHUS was poor with a 5-year death-censored graft survival of only approximately 50% with a 7% mortality.²² Graft failure was predominantly due to aHUS recurrence, which occurred in up to 70% of patients usually early in the course of the transplant (about 70% in the first year). The outcome of renal transplantation was largely predicted by the underlying genetic abnormality.²² In individuals with mutations in *CFH* the recurrence rate was >80%. Likewise activating mutations in *C3* and *CFB* had a high risk of renal recurrence. Unlike these fluid-phase complement proteins, *CD46* is membrane tethered and as such, a renal allograft will correct the complement defect and protect against aHUS. In keeping with this, the outcome following transplantation in those with mutations in *CD46* is much better with a very low recurrence rate. The posttransplant milieu [e.g., viral diseases, ischemia reperfusion injury, uncontrolled blood pressure, donor-specific antibodies, immunosuppressive calcineurin inhibitors (CNIs)] provides the necessary triggers to unmask the latent genetic complement defects by causing

endothelial cell damage and activation of the complement cascade. The CNIs (tacrolimus and cyclosporin), although consistently linked as a trigger for aHUS, have not been demonstrated to be significantly associated in two recent studies of aHUS recurrence posttransplant. Mammalian target of rapamycin inhibitors (e.g., sirolimus), however, have been reported to increase the risk of recurrence. Plasma therapy had a low success rate in rescuing recurrent aHUS following renal transplantation, although preemptive PEX has been associated with a trend of decreased recurrence. Such regimes have now been largely replaced by preemptive eculizumab at the time of transplantation.

Living-related kidney transplantation is not recommended in aHUS given that de novo aHUS has been recorded in four donors within a year of donation with complement mutations reported subsequently. Genotyping may reveal a known mutation in a family member; however, the presence of risk haplotypes and the fact that further genetic risk factors remain to be discovered make it impossible to rule out subsequent aHUS for a donor. If live related transplantation is to be considered, the donor should be counseled about the risk of de novo TMA.

DGKE THROMBOTIC MICROANGIOPATHY

DGKE-mediated aHUS is caused by homozygous or compound heterozygous mutations in this gene.¹⁹ Genetic pleiotropism, where a single gene affects a number of phenotypic traits in the same organism, is also seen in DGKE-mediated renal disease with a membranoproliferative glomerulonephritis (MPGN) phenotype also reported.²³

PATHOGENESIS

DGKE is an intracellular lipid kinase that converts diacylglycerol (DAG) to phosphatidic acid. DAG activates protein kinase C which leads to a number of downstream effects including the release of antithrombotic factors and prothrombotic factors as well as platelet activation, changes in vascular tone, and changes in the actin cytoskeleton. The ultimate mechanisms by which DGKE abnormalities result in TMA are yet to be elucidated. The disease appears to be independent of complement activation; however, rarely concomitant mutations in complement genes have been reported.

CLINICAL FEATURES

DGKE-mediated aHUS presents in infancy and early childhood. Heavy proteinuria is a predominant feature. The natural history of disease follows a relapsing/remitting course commonly progressing to chronic kidney disease (CKD) and ESRD.

TREATMENT

Treatment outcome data are limited with most cases progressing regardless of treatment. The disease appears to be complement-independent with reports of poor responses to eculizumab. Relapse of DGKE-mediated aHUS posttransplantation has not been reported.

METHYLMALONIC ACIDURIA AND HOMOCYSTINURIA, COBALAMIN C (CBL) TYPE HEMOLYTIC UREMIC SYNDROME

Methylmalonic aciduria and homocystinuria is a disorder of cobalamin (cbl; vitamin B₁₂) metabolism, which is associated with aHUS.²⁴ Homozygous or compound heterozygous mutations in the gene for the methylmalonic aciduria and homocystinuria type C protein, *MMACHC*, cause this disease. Cobalamin is a cofactor for both methionine synthase and methylmalonyl-coenzyme A mutase and its deficiency leads to methylmalonic acidemia with homocystinuria.

CLINICAL FEATURES

MMACHC-mediated aHUS predominantly presents in infancy or childhood, although rare cases in early adulthood have been reported. Plasma homocysteine levels are elevated and methionine levels are low. Methylmalonic acid is elevated in the urine. This metabolic disease has many extrarenal manifestations of variable severity (comprised development, ophthalmic, neurologic, and cardiac defects). Mortality is high if untreated or if there is cardiopulmonary involvement.²⁴

TREATMENT

Metabolic therapy with hydroxocobalamin and betaine is very effective. The role of complement is not clear; there are isolated reports of concomitant complement gene mutations and polymorphisms that may modify the disease, but the small number of published reports of eculizumab use describe nonresponse.

THROMBOTIC THROMBOCYTOPENIC PURPURA

INCIDENCE

The incidence of acquired TTP is 0.37 per 100,000 per year. The median age for diagnosis of the acquired form is 41 with 75% of cases occurring in female patients. Acquired TTP is very rare in children.²⁵

PATHOGENESIS

TTP is mediated by ADAMTS13 (A Disintegrin And Metalloprotease with a ThromboSpondin type 1 motif, member 13) deficiency, usually acquired with anti-ADAMTS13 autoantibodies (95%) and occasionally inherited with recessive mutations in the *ADAMTS13* gene. Ultra-large von Willebrand factor (ULvWF) multimers, which are synthesized and secreted by endothelial cells, are cleaved by ADAMTS13. In areas of very high shear stress (e.g., arterioles with high flow rates) these ULvWF multimers undergo a conformational change. In the absence of functional ADAMTS13, the ULvWF multimers accumulate on the endothelium to which platelets adhere. This results in the platelet-rich thrombi in the microvasculature which mediates end organ damage.²⁵ There is a genetic predisposition to the development of ADAMTS13

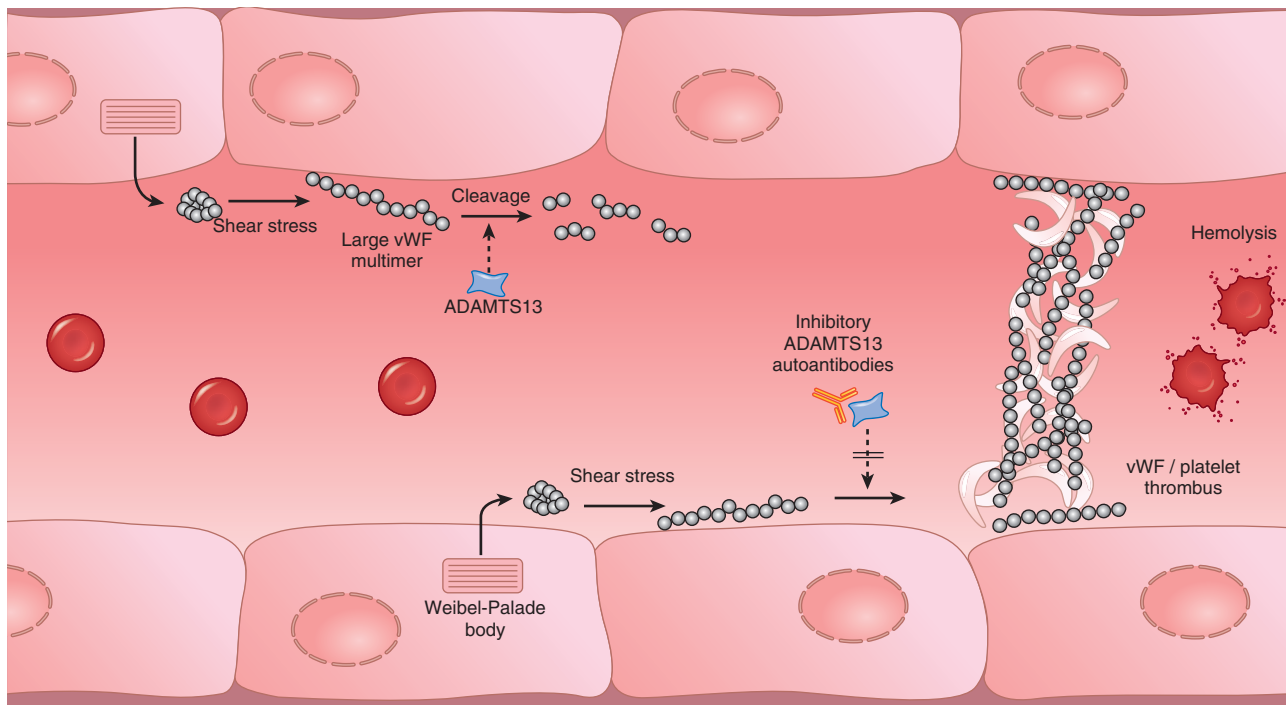


Fig. 34.7 Thrombotic thrombocytopenic purpura. Under normal circumstances ultra-large von Willebrand factor (vWF) multimers released from Weibel–Palade bodies in endothelial cells are cleaved by ADAMTS13 to prevent and regulate platelet adherence. In thrombotic thrombocytopenic purpura (TTP), ADAMTS13 deficiency, either acquired (ADAMTS13 autoantibodies) or inherited (recessive mutations in ADAMTS13), results in reduced cleavage of secreted or anchored ultra-large vWF strings. These form vWF–platelet thrombi, which result in tissue ischemia, platelet consumption, and microangiopathic hemolytic anemia. ADAMTS13, A disintegrin and metalloproteinase with a thrombospondin type 1 motif, member 13. (Modified from the National Renal Complement Therapeutics Centre 2017/2018 annual report. Available from: <http://www.atypicalhus.co.uk/>. Last accessed February 4, 2019.)

autoantibodies with HLA-DQ7, DRB1*11, and HLA-DRB3 linked in Caucasians. ADAMTS13 deficiency by itself may be insufficient to cause TTP in isolation, often requiring a secondary inflammatory or prothrombotic trigger (Fig. 34.7).

CLINICAL FEATURES

In comparison with complement-mediated aHUS, TTP is described to have less severe renal impairment (serum creatinine, 1.7–2.3 mg/dL) and more severe thrombocytopenia ($<30 \times 10^9/L$). Although a general guide to the likely cause of a TMA, these cut-offs are not absolute and cannot be relied upon in clinical practice where an ADAMTS13 activity is urgently required.²⁶

Although a potentially life-threatening disease the initial presenting clinical features may be mild and not of a critically ill patient. These symptoms may include nausea, vomiting, fatigue, dyspnea, bruising, and petechiae. Neurologic involvement is very common ranging from minor (e.g., headache, confusion, weakness) to major (seizure, stroke, coma) features. Gastrointestinal features are present in around 50%. Cardiac involvement can be seen with myocardial infarction. Arrhythmias are a frequent cause of death.²⁵

TREATMENT

TTP was historically almost universally fatal, but mortality decreased to approximately 10% following the introduction of PEX. Patients with TTP should be managed in a high

dependency area with monitoring. PEX should be initiated urgently.²⁷ Plasma infusion should only be used to temporize until the patient can be transferred to a facility able to undertake PEX. PEX is undertaken daily until recovery is seen. In refractory patients PEX may be increased to twice daily.

Immunosuppressive therapies (e.g., glucocorticoids and rituximab²⁸) are now standardly used to reduce the relapse rate in acquired TTP.

Relapse of acquired TTP was seen in around 33% of cases; however, with increasing use of rituximab this rate is falling. Around 50% of relapses will occur in the first year after presentation and routine clinical monitoring is required.

INFECTION-ASSOCIATED THROMBOTIC MICROANGIOPATHIES

SHIGA TOXIN HEMOLYTIC UREMIC SYNDROME

Shiga toxin–induced HUS is one of the main causes of AKI in young children and occurs following infection with Shiga toxin–producing enterohemorrhagic *Escherichia coli* (STEC) or *Shigella*.²⁹

MICROBIOLOGY

Shiga toxin–producing *E. coli* is the most common cause of HUS in the developed world. *E. coli* serotypes are classified by their O and H antigens. The O antigen is a polymer of

repeating oligosaccharides which forms the outermost domain of the lipopolysaccharide (LPS) in the outer membrane. The H antigen is a major component of bacterial flagella. In Europe and North America *E. coli* O157:H7 is the commonest serotype associated with STEC-HUS, although other non-O157 strains can cause disease (including O26, O80, O91, O103, O104, O111, O121, O145). In some regions non-O157:H7 strains predominate (e.g., Australia) and in May 2011 the largest recorded outbreak centered in Germany was with an *E. coli* O104:H4 strain. In Asia and Africa, *Shigella dysenteriae* serotype 1 is a major cause of HUS, again by the production of Shiga toxin.

SOURCE OF INFECTION

Shiga toxin-producing *E. coli* colonize healthy cattle intestine and also have been isolated from deer, sheep, goats, horses, dogs, birds, and flies. Meat, contaminated at slaughter, is the most common method of human infection although other vehicles for transmission have been described (Box 34.2), including unpasteurized dairy products, contaminated water sources, and infected vegetables.²⁹

S. dysenteriae 1 infection in the developed world is rare with contamination of food and water the usual sources. The infectious dose of *Shigella* is low and direct person-to-person spread is reported.

INCIDENCE

STEC-HUS accounts for over 90% of cases of HUS in children and occurs more commonly in summer months. In keeping with the mode of transmission, agricultural areas have a higher incidence. The incidence of STEC-HUS is 0.7 cases per 100,000/year predominantly in children under 5 years of age. HUS arises from around 5% to 10% of STEC infections, although this figure was much higher in the German O104:H4 outbreak (about 20%).³⁰ Less than 10% of children infected with *S. dysenteriae* 1 will develop a TMA.

CLINICAL FEATURES OF SHIGA TOXIN-PRODUCING ENTEROHEMORRHAGIC ESCHERICHIA COLI-HEMOLYTIC UREMIC SYNDROME

Bloody diarrhea is the classical prodromal feature of a STEC-HUS. Three days postexposure to STEC, diarrhea will develop in association with vomiting and abdominal pain. In the 5% to 10% of patients who develop HUS, this will occur around 7 days later. Although conventionally reported as bloody, this is only reported in approximately 60% of cases (Fig. 34.8). It is not clear why some patients progress and others do not.

In around 5% of cases of STEC-HUS, no diarrhea is reported and thus investigations for STEC should be carried out in all cases of TMA regardless of the absence of diarrhea.³¹ Urinary tract infections with STEC have been documented to cause HUS. A high white cell count is common in STEC-HUS and correlated with a poorer outcome. Over 50% of STEC-HUS cases will require dialysis, frequently in those with concurrent decreased effective arterial volume at admission. Despite this, renal recovery is the rule in STEC-HUS with dialysis dependency for more than 2 weeks being unusual. Hypertension is also frequently observed. Mortality in STEC-

Box 34.2 Vehicles for Transmission of *Escherichia coli* O157:H7

Meat

- Undercooked ground meat
- Undercooked steak
- Undercooked roast beef
- Salami
- Deer jerky
- Venison

Contaminated Fruit/Vegetables

- Lettuce
- Bean sprouts
- Radish sprouts
- Salad
- Melons
- Coleslaw
- Grapes
- Unpasteurized apple juice/cider

Dairy Products

- Unpasteurized milk
- Cheese curd from unpasteurized milk
- Butter from unpasteurized milk
- Ice cream bars

Water

- Lakes/ponds
- Swimming pools
- Municipal drinking water
- Other

Fecal-Oral

- Daycare centers

Airborne

HUS is low at approximately 2% and occurs predominantly in the very old and very young.²⁹

EXTRARENAL MANIFESTATIONS

Extrarenal manifestations in STEC-HUS are common. Neurological symptoms are commonly reported and these may range from headaches to coma and cortical blindness. Gut effects such as hemorrhagic colitis, bowel necrosis, perforation, and intussusception are seen with longer-term colonic strictures reported. Pancreatitis is seen in STEC-HUS and impaired glucose tolerance and diabetes mellitus may be develop (Box 34.1).

LONG-TERM SEQUELAE

Although ESRF is unusual, impaired renal function, proteinuria, and hypertension have been reported in up to 40%. Less commonly ongoing diabetes mellitus and neurological defects and seizures may be seen.³²

CLINICAL FEATURES OF SHIGELLA DYSENTERIAE TYPE 1

Shigella typically presents with a similar prodrome to STEC: diarrhea (initially watery, subsequently mucoid, or bloody), abdominal pain, and vomiting. Fever is common and an

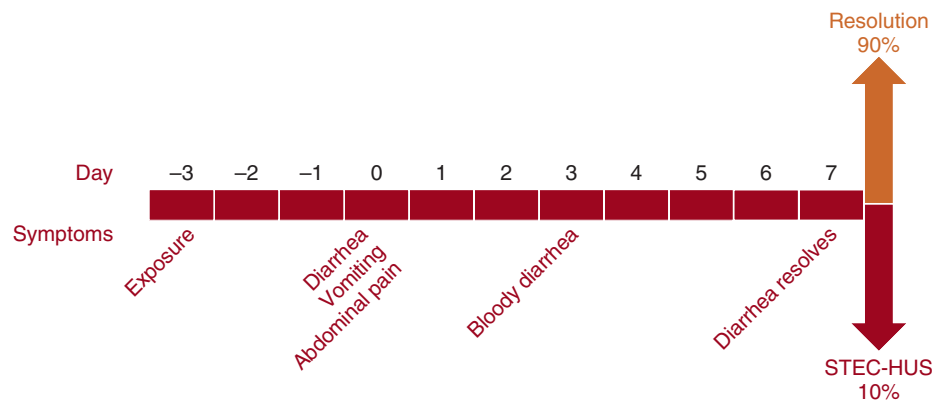


Fig. 34.8 Clinical time line of *Escherichia coli* O157 infection. Three days postexposure to Shiga toxin-producing *Escherichia coli* (STEC), diarrhea will develop in association with vomiting and abdominal pain. Thus around day 3 bloody diarrhea may be seen. In about 10% that develop hemolytic uremic syndrome (HUS), this will occur around 7 days later.

elevated white cell count may be seen. On average, symptoms develop 3 days after infection. In comparison with STEC-HUS the disease has a poorer outlook with 15% mortality and 40% of patients developing CKD. The presentation with HUS is at about day 7 when the diarrheal illness has resolved. As with STEC-HUS, gastrointestinal and neurological manifestations can arise.

MICROBIOLOGY TESTING

Stool samples should be sent in all cases of TMA because approximately 5% of patients lack a diarrheal prodrome. Diarrhea may have stopped by the time of presentation, but STEC can still be cultured from feces or rectal swab. Culture is not a sensitive test, especially if sample collection is delayed as bacteria are usually only present for a few days and stool samples are not routinely processed after many days in hospital, especially when diarrhea has stopped. In addition, polymerase chain reaction testing for Shiga toxin 1/2 genes or enzyme-linked immunosorbent assay (ELISA) measurement to detect Shiga toxin 1/2 in the stools is now recommended. Serological detection of anti-LPS antibodies against *E. coli* (immunoglobulin M [IgM] positive/rising IgG titer) may also suggest infection.

PATHOGENESIS

STEC reaches the gut and closely adheres to the epithelial cells of the gastrointestinal mucosa through a 97-kD outer membrane protein, intimin, resulting in the formation of effacement lesions. The intimin protein is encoded by the *eaeA* gene in the locus of enterocyte effacement. Following this attachment there is translocation of toxin (usually Shiga toxin type 2) through the epithelium into the circulation. The toxin is thought to circulate attached to leucocytes, erythrocytes, and platelets. Shiga toxins consist of an A unit and five B units. The pentameric B unit binds to globotriaosyl ceramide (Gb3) on expressing organs mediating internalization in clathrin-coated pits and retrograde transport of the toxin via the Golgi complex to the endoplasmic reticulum. The A unit inhibits ribosomal function and protein synthesis, leading to endothelial cell death and exposure of the underlying basement membrane.²⁹ In addition, Shiga toxin is able to enhance

the release of proinflammatory cytokines (interleukin-1 [IL-1], tumor necrosis factor- α , and IL-6), amplifying the inflammatory events and contributing to the procoagulant milieu and ultimately a TMA.³³ Although Gb3 is found in other organs, the kidney is most susceptible to injury. This may be due to higher levels of Gb3 on renal cells, particularly the glomerular endothelium (but also podocytes, mesangial, and tubular epithelial cells), the high blood flow to the kidney, or a greater susceptibility of renal cells to the effects of the toxin. The expression of Gb3 at other sites (e.g., brain) may explain extrarenal manifestations of STEC-HUS. Increased expression of the Gb3 receptor in children has also been proposed as a cause for the predominant pediatric presentation, although other explanations have also been proposed (e.g., increased prevalence of anti-Shiga antibodies in adults; Fig. 34.9).

ROLE OF COMPLEMENT IN THE PATHOGENESIS OF STEC-HUS

Complement activation, particularly the alternative complement pathway, has long been implicated in the pathogenesis of STEC-HUS with the frequent finding of low C3 levels and increased breakdown products of C3 and FB suggesting activation of the AP of complement.³⁴ The presence of low C3 is associated with up to 50% of cases of STEC-HUS and it has been reported that persistently low levels are associated with a poorer prognosis. In vitro studies have demonstrated that STEC induces P-selectin expression at the membrane surface, increasing complement AP activation. There is evidence from a preclinical rodent model of Shiga toxin-mediated injury that complement activation may augment the effects of the toxin on cells, although no evidence of complement activation was seen in a primate model. It remains uncertain whether this complement activation plays a role in the development of STEC-HUS and whether complement blockade may improve outcome. In rare cases functionally significant complement mutations are also present and often result in a poorer prognosis.³⁵

TREATMENT

As a self-limiting condition, supportive management with fluid resuscitation, blood pressure control, and dialysis remain

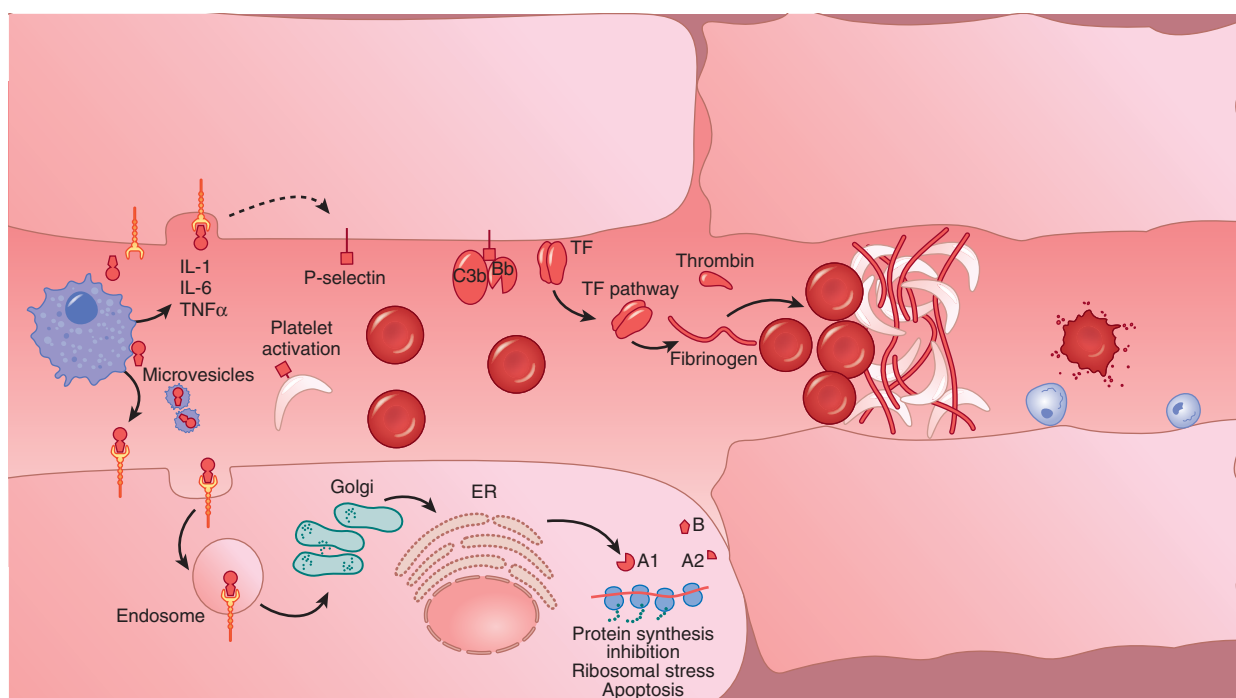


Fig. 34.9 Shiga toxin hemolytic uremic syndrome. Shiga toxin is transported from the gut to the kidney via leucocytes, erythrocytes, and platelets. Toxin released in the circulation or within microvesicles is internalized in clathrin-coated pits and thereafter undergoes retrograde transport via the Golgi complex to the endoplasmic reticulum. The A unit of Shiga toxin inhibits ribosomal function and protein synthesis, leading to endothelial cell death and exposure of the underlying basement membrane. Shiga toxin is also able to enhance the release of proinflammatory cytokines [interleukin-1 (IL-1), tumor necrosis factor- α (TNF α), and IL-6], thus amplifying the inflammatory events. Shiga toxin can also upregulate P-selectin and cause complement activation. The consequent thrombosis results in microangiopathic hemolytic anemia and end organ damage. (Modified from the National Renal Complement Therapeutics Centre 2017/2018 annual report. Available from: <http://www.atypicalhus.co.uk/>. Last accessed February 4, 2019.)

the mainstay of treatment for STEC-HUS. Antimotility agents should be avoided because they increase the risk of HUS. The use of antibiotics is controversial with most centers avoiding their use in the treatment of *E. coli* O157 infection because of either no benefit or potential increase in the risk of developing HUS, although this may be dependent on the antibiotic used. Some classes of antibiotics (e.g., fluoroquinolones) dramatically increase the production and the release of Shiga toxin before bacterial lysis and worsen the outcome of STEC-HUS in animal models. Bacteriostatic agents (e.g., azithromycin) block protein synthesis and have a strong inhibitory effect on Shiga toxin production and release by STEC. A randomized controlled trial of azithromycin in STEC-HUS (NCT02336516) is underway.

By contrast, antibiotic treatment in *S. dysenteriae* type 1 has not been associated with increased HUS risk, possibly due to the differences in control of Shiga toxin production.

Although there is evidence of complement activation in STEC-HUS, the role of the complement inhibitor eculizumab remains unproven. Several case reports in 2011 described renal and neurological recovery in children treated with eculizumab for STEC-HUS.³⁶ Subsequently, in the 2011 O104:H4 HUS outbreak in Europe, retrospective analyses did not demonstrate a beneficial role of eculizumab or PEX over supportive care, although the patients who were treated with eculizumab had more severe disease.³⁰ Ultimately in a self-limiting disease a randomized clinical trial of eculizumab for STEC-HUS will be required to define the role of complement inhibition and these are underway (ECUSTEC).

Because patients usually shed STEC for a week, standard precautions to prevent person-to-person transmission and source isolation should be implemented. In children it is usual to have at least two negative stool cultures prior to returning to school.

PNEUMOCOCCAL HEMOLYTIC UREMIC SYNDROME

TMA may occur in adults and children in the context of invasive *Streptococcus pneumoniae* infection.³⁷

INCIDENCE

Pneumococcal HUS has been reported to have a 10-year cumulative incidence rate of 1.2 per 100,000 children. It complicates 0.5% of pneumococcal disease, usually in those with pneumonia and empyema.³⁷

PATHOGENESIS

Several mechanisms have been suggested to explain the pathogenesis of pneumococcal HUS. Pneumococci produce neuraminidase, which cleaves sialic residues from glycoproteins on erythrocyte, platelet, and endothelial cell membranes, exposing the cryptic Thomsen–Friedenreich antigen (T antigen). Natural IgM antibodies can then bind and result in cell damage and a TMA.³⁸ The DAT (Coombs) test is thus

positive. The cleavage of sialic acid has also been hypothesized to reduce FH binding, resulting in impaired endothelial complement regulation, thus contributing to disease pathogenesis.³⁹ The pneumococcal protein PspC has also been demonstrated to bind plasminogen and, when converted to plasmin, damage the endothelium, again contributing to the disease mechanism. Again, as with STEC-HUS, rare genetic variants in the complement system are only rarely found.

CLINICAL FEATURES

Most pneumococcal HUS is associated with pneumonia (approximately 70%), frequently with empyema. Meningitis (around 20%), sinusitis, and otitis media have also been associated with HUS. Infected children are usually extremely unwell and frequently require prolonged hospitalization and intensive care unit admission. The majority (approximately 75%) require dialysis. The mortality is much higher than seen in STEC-HUS (around 10%) usually due to the underlying infection. Up to around 33% develop ESRD. As with all other TMAs extrarenal manifestations have been reported.⁴⁰

TREATMENT

Supportive management and treatment of the infection should be the focus. The role of PEX is controversial with a theoretical risk of worsening the TMA with donor plasma because it may contain anti-T IgM. The role of eculizumab in the treatment of pneumococcal HUS is unclear with only sporadic case reports documenting response.¹⁷

HUMAN IMMUNODEFICIENCY VIRUS-ASSOCIATED THROMBOTIC MICROANGIOPATHIES

Human immunodeficiency virus (HIV)-associated TMA was common in the pre-highly active antiretroviral therapy era with reported incidences from 1.5% to 7%. With the currently available treatments for HIV this has fallen to approximately 0.3%. The pathogenic mechanisms are unclear but are thought to reflect glomerular endothelial damage. HIV TMAs are associated with lower CD4⁺ cell counts, higher viral RNA levels, and opportunistic infections. Treatment consists of supportive care and antiretroviral treatment.⁴¹

OTHER INFECTIONS

In addition to *E. coli*, HIV, and *S. pneumoniae*, a large number of other bacterial, viral, fungal, and parasitic infections have been described in association with TMA (Box 34.3). Unlike *E. coli* and *S. pneumoniae* it is unclear if this is a direct effect of the pathogen, a side effect of treatment, or a trigger that unmasks a latent complement defect (Fig. 34.6). In studies of complement-mediated aHUS in those with *CFH*, *CFI*, or *CD46* mutations 70% were reported to have an infectious trigger. Supportive care and treatment of the infection are recommended and complement evaluation should be undertaken. Eculizumab can be considered when the infectious agent is considered a trigger of complement-mediated aHUS.¹

Box 34.3 Infections Linked to Hemolytic Uremic Syndrome

Shiga Toxin Diarrheal Illnesses

Escherichia coli O157:H7 O26, O80, O91, O103, O104, O111, O121, O145
Shigella dysenteriae type 1

Non-Shiga Toxin Diarrheal Illnesses

Norovirus
Campylobacter upsaliensis
Clostridium difficile
Respiratory infections
Bordetella pertussis infection
Streptococcus pneumoniae
Haemophilus influenzae

Other Bacterial

Fusobacterium necrophorum

Viral Illnesses

Varicella
Cytomegalovirus
Influenza H1N1
Hepatitis A
Hepatitis C
Human immunodeficiency virus
Coxsackie B virus
Epstein-Barr virus
Dengue
Human herpesvirus 6
Parvovirus B19

Parasites

Plasmodium falciparum

SECONDARY THROMBOTIC MICROANGIOPATHIES

PREGNANCY-ASSOCIATED THROMBOTIC MICROANGIOPATHIES

Pregnancy-associated TMAs include primary complement-mediated aHUS and TTP. Additionally, there is a broader differential diagnosis which includes preeclampsia, HELLP (hemolysis, elevated liver enzymes, and low platelets), fatty liver, placental abruption, and postpartum hemorrhage (Fig. 34.10). See Chapter 48 for a more detailed discussion of pregnancy-related renal disorders.

Pregnancy is a common trigger of aHUS in women, accounting for 20% of cases, usually in the postpartum.⁴² It has become clear that a high proportion will have identifiable complement mutations and that pregnancy acts as a trigger in those with an underlying genetic predisposition. Historically outcomes were poor with approximately 75% developing ESRD despite PEX. The high frequency of complement mutations in pregnancy-associated aHUS provides the rationale for complement inhibition with eculizumab, although no clinical trials have been performed in this setting.⁴² Many reports about the safe use of eculizumab in

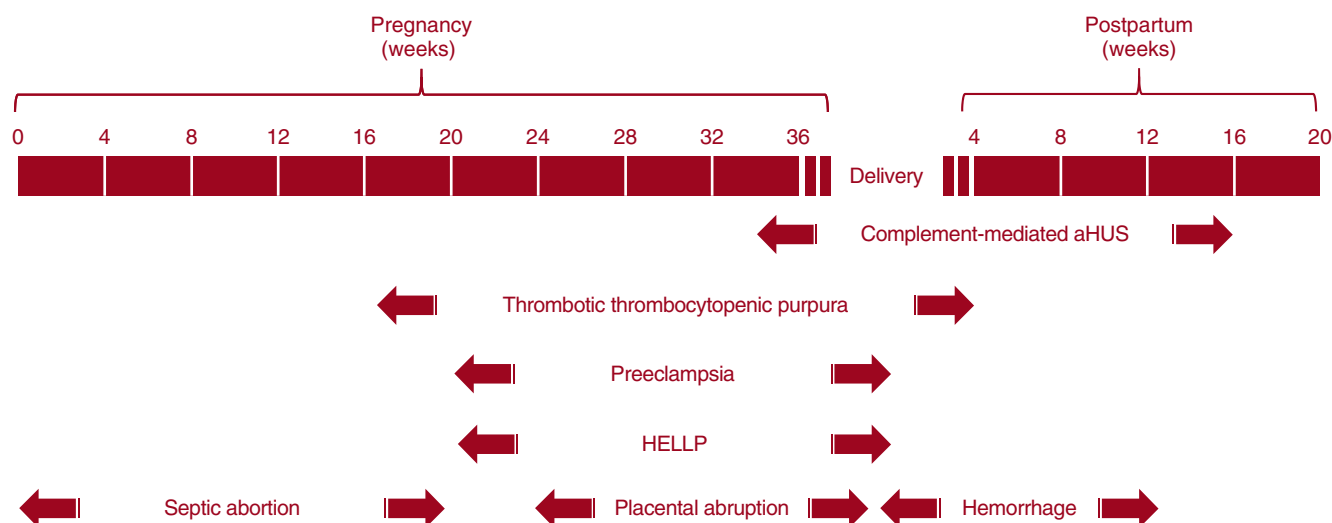


Fig. 34.10 Presentation and differential diagnosis of thrombotic microangiopathies (TMAs) during pregnancy and postpartum. Timing of presentation can help with the differential diagnosis of TMA, with complement-mediated atypical hemolytic uremic syndrome typically presenting in the postpartum. *aHUS*, Atypical hemolytic uremic syndrome; *HELLP*, hemolysis, elevated liver enzymes, and low platelets.

pregnancy for paroxysmal nocturnal hemoglobinuria and aHUS have now been published.

A significant proportion of women present with TTP during the second and third trimesters of pregnancy.⁴³ The physiological increase in vWF during pregnancy, which consumes ADAMTS13, has been proposed as a predisposition and in women with a genetic predisposition its activity can fall low enough for TMA to manifest. PEX is recommended.

Preeclampsia is characterized by new-onset hypertension and proteinuria or other systemic signs (e.g., thrombocytopenia and AKI). Preeclampsia is the most common cause of AKI in pregnancy, although dialysis requirement is rarely required. Preeclampsia complicates up to 5% of all pregnancies and hypertension, diabetes mellitus, and CKD increase the risk.⁴³ HELLP is thought to be a severe atypical form of preeclampsia. It complicates 10% to 20% of preeclampsia cases and about 0.1% of all pregnancies. The pathogenesis of preeclampsia and HELLP is not well defined. Increased circulating levels of the syncytiotrophoblast-derived antiangiogenic factors, soluble endoglin and the soluble form of the vascular endothelial growth factor receptor (sFlt1/soluble VEGFR1), may contribute to the observed endothelial dysfunction. Although it is often difficult to differentiate clinically between HELLP and TMA, in HELLP, the predominant glomerular pathology is endotheliosis. In contrast to pregnancy-associated aHUS, only a minority (8%–10%) of patients with preeclampsia and HELLP syndrome harbor complement gene variants, mostly of unknown or nonpathogenic significance.⁴⁴ Management is supportive, with expedited delivery being the definitive treatment for HELLP. Expectant management may be considered if the woman is not at or near term. While there is some evidence of complement activation in HELLP and preeclampsia, its role in the pathogenesis of disease is uncertain. It should be noted that in women receiving eculizumab for paroxysmal nocturnal hemoglobinuria⁴⁵ or complement-mediated aHUS, preeclampsia has occurred, arguing against a role for C5 inhibition in the management of preeclampsia.

DRUG-MEDIATED THROMBOTIC MICROANGIOPATHY

Although TMA has been linked to a large number of drugs, causality has only been established in relatively few (Box 34.4). Drug-mediated TMA occurs by two main mechanisms: direct toxicity and immune-mediated damage. For example, interferon- β ⁴⁶ and bevacizumab⁴⁷ cause TMAs by a dose-dependent toxicity, whereas quinine induces the development of autoantibodies reactive with either platelet glycoprotein Ib/IX or IIb/IIIa complexes, or both.⁴⁸ Supportive care and discontinuation of the causative drug are the standard treatment options with the exception of ticlopidine which has been associated with anti-ADAMTS13 antibodies and TTP, and therefore PEX is recommended.

MALIGNANCY-ASSOCIATED THROMBOTIC MICROANGIOPATHY

There are many reports of malignancy-associated TMAs in the literature.³ The pathogenesis remains obscure. However, erythrocyte shearing after direct contact with microvascular embolic tumor cells has been suggested as a possible mechanism. It is frequently difficult to discriminate between a direct effect of the malignancy and chemotherapy-mediated effect in the cause of the TMA. Prognosis is poor because of malignancy-related mortality. Withdrawal of chemotherapy and supportive management are usually initiated.

DE NOVO THROMBOTIC MICROANGIOPATHY AFTER SOLID ORGAN TRANSPLANT

De novo TMAs following solid organ transplants (liver, pancreas, lung, and heart) are reported. The pathogenesis

Box 34.4 Drugs Associated With Thrombotic Microangiopathies

Cisplatin
Gemcitabine
Vincristine
Mitomycin
Clopidogrel
Quinine^a
Interferon α
Interferon β^a
Oxaliplatin
Penicillin
Pentostatin
Quetiapine
Sulfisoxazole
Sunitinib
Ticlopidine
Muromonab-CD3
Antivascular endothelial growth factor bevacizumab^a
Campath
Cyclosporin
Tacrolimus
Everolimus
Sirolimus
Ciprofloxacin
Oral contraceptives
Illicit drugs (e.g., cocaine, heroin, ecstasy)

^aAlthough many drugs have been reported in association with thrombotic microangiopathy, compelling evidence for causality has been demonstrated in very few, including antivascular endothelial growth factors, interferon B, and quinine. (Modified from George JN. Interferon-induced thrombotic microangiopathy. *Blood*. 2016;128:2753–2754.)

is multifactorial and is likely to reflect the endothelium-damaging milieu seen posttransplant, including immunosuppressant drugs such as CNIs, ischemia reperfusion injury, antibody-mediated rejection, viral infections (e.g., cytomegalovirus), etc. In one series of de novo TMA after kidney transplantation, complement mutations were identified in about 30%.⁴⁹ Thus eculizumab therapy should be considered if the primary diagnosis is not incompatible with complement-mediated aHUS. Supportive treatment and removal of the precipitating factors (CNI discontinuation or dose reduction, treatment of antibody-mediated rejection, and viral infections) may be sufficient to stop the TMA.

THROMBOTIC MICROANGIOPATHY AFTER BONE MARROW TRANSPLANT

TMA is reported to complicate 10% to 40% of allogeneic bone marrow transplants with mortality rates reported up to 75%. As with TMA, following solid organ transplantation there is a multifactorial pathogenesis including CNIs, graft versus host disease, HLA mismatch, chemotherapy, radiation therapy, and infections. FH autoantibodies and rare genetic variants in aHUS-associated genes have rarely been reported in postbone marrow transplant TMA.⁵⁰

The optimal therapeutic strategy has yet to be defined for this treatment group. The occasional reports of complement

mutations and evidence of complement activation have provoked the use of eculizumab with uncontrolled retrospective analyses describing response. Prospective trials are likely needed in order to establish the optimal strategy.

AUTOIMMUNE AND GLOMERULAR DISEASE-ASSOCIATED THROMBOTIC MICROANGIOPATHIES

TMA has been reported in association with many glomerular diseases: MPGN/C3 glomerulopathy (C3G), focal segmental glomerulosclerosis (FSGS), IgA nephropathy, antineutrophil cytoplasmic antibody-associated vasculitis, membranous nephropathy. Frequently this is a histopathological description without hematological or biochemical features.

In MPGN/C3G, acquired and hereditary complement abnormalities are seen (Chapter 31). These are similar (mutations in *CFH* and *C3*) although subtly different to those seen in complement-mediated aHUS, and concurrent and sequential manifestation of C3G and TMA has been reported. This genetic pleiotropy is poorly understood, and may reflect the location of mutations within these genes and also genetic modifiers of disease. Genetic pleiotropy has also been reported in *DGKE*- and *INF2*-mediated disease with TMA reported in addition to the MPGN or FSGS, respectively.

Autoimmune diseases such as systemic lupus erythematosus (SLE), catastrophic antiphospholipid syndrome (CAPS), and scleroderma renal crisis (SRC) can also present with a TMA. The exact pathogenesis has yet to be defined; however, in CAPS and SLE there is evidence of complement activation. For SLE, CAPS, and SRC with TMA, treatment should be of the underlying condition.

For SRC the use of angiotensin-converting enzyme inhibitors has been demonstrated to significantly reduce mortality.

In SLE associated with a TMA there is no evidence to suggest additional therapy other than standard lupus management. It is not clear whether a TMA in the setting of SLE worsens the outcome.

In CAPS, the incidence of TMA is about 14% with an overall mortality of 37%. Anticoagulation with glucocorticoids with or without PEX is the most common treatment. In CAPS, there is evidence of complement activation in both clinical setting and mouse models. There are case reports of successful use of eculizumab in CAPS and prospective trials are underway.

SEVERE HYPERTENSION-ASSOCIATED THROMBOTIC MICROANGIOPATHY

Severe hypertension can present with a phenotype indistinguishable from complement-mediated aHUS in the acute setting. In practice, failure of blood pressure control and supportive management to control the TMA will often result in the pragmatic initiation of PEX or eculizumab until complement evaluation is available. In the majority of patients with TMA associated with severe hypertension, renal function and MAHA usually recover with management of blood pressure. In a retrospective case series, genetic analysis identified rare variants in complement genes in patients for whom

TMA was initially attributed to severe hypertension; eight of nine patients progressed to ESRD despite management of hypertension. This may be particularly relevant when considering kidney transplant assessment in an individual with ESRD attributed to severe hypertension with TMA, or if TMA occurs again after transplantation.

SUMMARY

TMA can manifest in a wide range of conditions and can be associated with significant morbidity and mortality. The different pathogenesis of these TMAs mean that personalized management of patients is required to provide the targeted therapy best suited to the underlying cause.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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